

IT3051 – FUNDAMENTALS OF DATA MINING

MINI PROJECT ASSIGNMENT 2026

1. Project Overview

This group mini project requires students to apply the complete data mining and machine learning workflow to a real-world problem. Each group should identify a specific problem scenario. Students are responsible for identifying a suitable publicly available dataset and obtaining instructor validation before continuing with the project.

The project should progress from problem understanding and dataset selection through data preparation, exploratory analysis, model development, optimization, and implementation of a user-friendly prediction system.

2. Assessment Structure

Component	Weight	Assessment Type
Progress Evaluation 1	30%	Individual evaluation
Progress Evaluation 2	30%	Individual evaluation
Final Presentation & System Demonstration	20%	Group presentation, individually assessed
Technical Report	20%	Group evaluation

Total: 100%

More details about each evaluation will be shared.

3. Project Workflow and Requirements

Stage 1 – Understand the Assigned Problem Scenario

Each group should identify a real-world problem/scenario. Clearly describe the problem to be solved.

- Identify the prediction objective and expected target variable.
- Identify potential users/stakeholders and the decisions the system could support.
- Identify important user requirements and expected system inputs/outputs.
- Determine whether the problem is a classification, regression, or another suitable data mining task.
- Do not select a dataset simply because it is easy to use; the dataset must support the identified scenario.

Stage 2 – Dataset Identification and Instructor Validation

Students must independently search for and select a suitable publicly available dataset. The selected dataset must be validated by the instructor, before EDA, preprocessing, model training, or system development is carried out. Students should get the dataset validated during the lab sessions. Prepare a dataset proposal and show the proposal and the dataset to a instructor and get it approved.

The dataset proposal must include:

- Dataset name and original source.
- Dataset URL / download source and proper citation.
- Dataset description and real-world context.

- Number of records and relevant features.
- Target variable and its meaning.
- Expected machine learning task (e.g., classification or regression).
- Reason why the dataset is suitable for the assigned problem.
- Important data-quality observations or known limitations.
- Potential ethical, privacy, licensing, or accessibility considerations, if applicable.

Important: Students must NOT proceed with project development using an unvalidated dataset. If the proposed dataset is rejected, an alternative dataset must be identified and submitted.

Dataset Validations Deadline: [20th September 2026] - Submit the proposal to the submission link, before you come for the validation, For KandyUni and Northern Uni Students the dates will be informed via the staff members.

Stage 3 – Data Understanding and Exploratory Data Analysis (EDA)

After dataset approval, students must perform a detailed investigation of the data. You should investigate through below given aspects, but **MUST NOT** limit only to given aspects.

- Understand the structure, meaning, and distribution of the data.
- Identify variable types and relevant relationships.
- Analyse missing values and duplicated records.
- Investigate outliers and unusual observations.
- Examine class imbalance where applicable.
- Visualise important distributions and relationships.
- Identify potential data leakage and check how it will be prevented.
- Document observations that influence later preprocessing and modelling decisions.

Stage 4 – Data Preprocessing and Feature Engineering

Students are expected to perform appropriate preprocessing based on the selected dataset. You can do preprocessing and feature engineering through below given aspects, but **MUST NOT** limit only to given aspects.

- Handle missing values appropriately.
- Handle duplicate or invalid records where necessary.
- Treat outliers where justified.
- Encode categorical variables appropriately.
- Scale/normalize numerical variables when required by the selected algorithms.
- Perform appropriate feature engineering.
- Perform feature selection where appropriate.
- Separate training and testing data appropriately and avoid information leakage.

Students should be able to justify why each important preprocessing decision was made.

Stage 5 – Progress Evaluation 1 (30%)

This is an individual evaluation. A viva will be conducted during the lab, assessing each group member where each student must demonstrate their own understanding of the work completed up to the preprocessing stage.

- Problem/scenario understanding, Dataset selection and justification, Dataset characteristics and target variable, EDA findings, Data-quality issues, Preprocessing techniques and reasons for using them, Feature engineering/selection decisions, Data leakage risks and prevention, Ability to explain the student's own contribution, etc.

Evaluation Dates: [23rd, 26th, 27th September], For KandyUni and Northern Uni Students the dates will be informed via the staff members.

Stage 6 – Model Development

Students must develop and compare multiple suitable machine learning models rather than relying on a single algorithm.

- Implement at least FOUR suitable machine learning algorithms.
- Select algorithms appropriate to the problem and dataset.
- Use an appropriate validation strategy.
- Evaluate models using suitable performance metrics.
- Compare model performance systematically.
- Record experiments, results, and important observations.
- Develop the understanding why certain models perform better or worse.

Stage 7 – Model Optimization and Final Model Selection

- Perform hyperparameter tuning for suitable models.
- Use an appropriate optimization/search strategy.
- Investigate whether feature selection/engineering or other modelling choices improve performance.
- Compare tuned models with baseline models.
- Select the final model.
- Clearly justify the final model selection.

Stage 8 – Progress Evaluation 2 (30%)

This is an individual evaluation focused on the modelling and optimization stages.

- Algorithm selection and justification, Model comparison and evaluation metrics, Validation strategy, Hyperparameter tuning, Optimization decisions, Interpretation of results, Final model selection and justification, Ability to explain the student's own contribution and technical decisions, etc.

Evaluation Dates:[30th September, 3rd and 4th October], For KandyUni and Northern Uni Students the dates will be informed via the staff members.

Stage 9 – Backend Development

The final trained model must be integrated into a functional backend/service that can receive user inputs and return predictions.

- Load the final trained model and required preprocessing pipeline.
- Accept and validate user inputs.
- Apply the same preprocessing used during model development.
- Generate predictions using the final model.
- Return clear and meaningful prediction results.
- Handle invalid or missing inputs appropriately.

Stage 10 – Frontend Development and User Experience

Develop a user-friendly frontend that allows the intended user to interact with the prediction system.

- Provide clear input fields and instructions.
- Validate user inputs.

- Present prediction results clearly.
- Follow the requirements of the assigned scenario.
- Provide a simple and understandable user experience.
- Ensure the frontend and backend work together as an end-to-end system.

Stage 11 – Technical Report (20%)

The group must submit a technical report covering the complete project.

- Introduction and problem definition.
- Assigned scenario and user/stakeholder requirements.
- Dataset identification, source, citation, and validation.
- Data understanding and EDA.
- Data cleaning and preprocessing.
- Feature engineering and feature selection.
- Algorithms implemented and their rationale.
- Model evaluation and comparison.
- Hyperparameter tuning and optimization.
- Final model selection and justification.
- System architecture and implementation.
- Backend and frontend development.
- System testing and results.
- Limitations and possible future improvements.
- Individual/group contributions, where required.

Final Report Submission Deadline: [6th October], *For KandyUni and Northern Uni Students the dates will be informed via the staff members.*

Stage 12 – Final Presentation and System Demonstration (20%)

The final presentation will be conducted as a group presentation; however, students will be assessed individually.

Students need to physically present their completed data mining solution as if they are presenting it to non-technical stakeholders or decision-makers within a company.

The focus is on their ability to translate technical findings into clear business insights and actionable recommendations, rather than simply explaining the algorithms or code. The slides for the presentation should also be made appropriately.

Students should demonstrate that they can:

- Clearly explain the business problem and why it matters to the organisation.
- Explain the solution and its purpose in simple, non-technical language.
- Present the key findings, patterns, predictions, or insights obtained from their analysis.
- Explain how the solution can support business decisions or improve the organisation's operations.
- Demonstrate their system/application and explain its functionality from a user/business perspective.
- Justify their recommendations using evidence from their analysis.
- Communicate effectively with a non-technical audience, avoiding unnecessary technical terminology.
- Respond confidently to questions from the audience and justify their individual contribution and decisions.

Final Submission Deadline for all the deliverables : [6th October], *For KandyUni and Northern Uni Students the dates will be informed via the staff members.*

Final Presentation Date: [7th, 10th, 11th October], *For KandyUni and Northern Uni Students the dates will be informed via the staff members.*

4. Dataset Selection Guidelines

A suitable dataset should:

- Be directly relevant to the assigned problem scenario.
- Contain a clearly identifiable target variable suitable for prediction.
- Contain sufficient observations and useful predictive features.
- Come from a credible and publicly accessible source.
- Be accessible to all group members for academic use.
- Contain enough complexity to support meaningful EDA, preprocessing, modelling, and optimization.
- Allow the group to demonstrate appropriate data mining concepts.
- Have no unnecessary privacy, legal, or ethical concerns.

Possible sources may include UCI Machine Learning Repository, Kaggle, government open-data portals, research datasets, and other credible public data repositories. Students should cite the original source of the data wherever possible.

5. Important Technical Requirements

- The dataset must be validated by the instructor before project development proceeds.
- At least FOUR machine learning algorithms must be implemented and compared.
- Appropriate evaluation metrics must be selected according to the validated problem and dataset.
- All major technical decisions must be justified using evidence from the data and experiments.
- The final system must use the final selected model and the correct preprocessing pipeline.
- Each student must understand the complete project sufficiently to answer individual evaluation questions.
- Reusing another group's dataset, code, model, or results without permission/acknowledgement is not acceptable.

6. Required Final Deliverables

- Approved dataset and dataset source/citation.
- Dataset proposal/validation evidence.
- Source code for data pre-processing and model development.
- Trained final model.
- Source codes of the final integrated system.
- Technical report.
- Final presentation materials.
- Any required experiment/evaluation evidence